Benefits and Challenges in Adoption of Agriculture Drones in India

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1. INTRODUCTION

In recent years, drone technology has become increasingly popular in the agricultural sector. Drone provides many benefits to farmers, including increased efficiency, better yields and operational costs. However, there are concerns that farmers may be reluctant to adopt drone technology due to the possibility of job loss or lack of knowledge and training. We can find out the benefits of drone technology and challenges in the agricultural sector that can withdraw farmers from adopting this technology.

In recent times, several announcements by the Government of India have focused the government on promoting drones in the agriculture sector. According to the Indian federal budget of 2022-23, (Anonymous. 2023b) the government is eager to use the word 'Kisan' drone to understand the farmer as well as promote the agricultural sector in the country. Farmer drones are being promoted for crop evaluation, digitization of land records and use of pesticides and nutrients. Some specific drone applications are pesticides, soil and crop nutritional spraying, soil samples, farm animal monitoring,

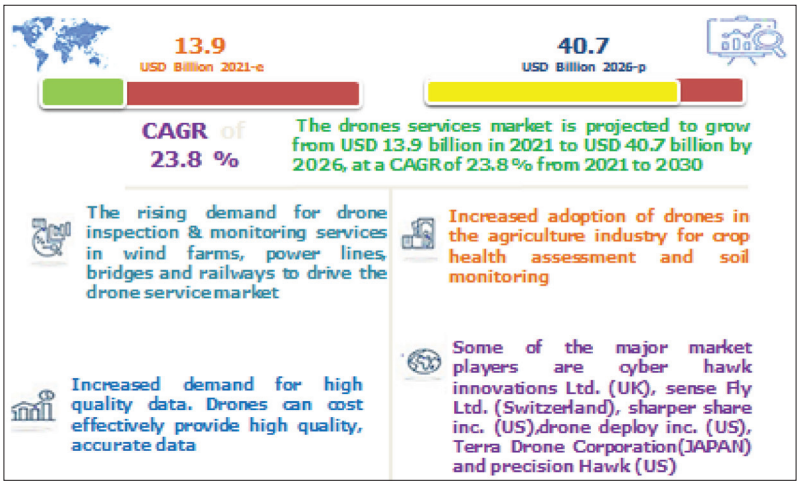

Fig. 1: Attractive Growth opportunities in Drone Service Market (Anonymous, 2022)

real-time aerial imagination and sensor data collection, yield estimate, etc. (Costopolo C. *et al*., 2022).

In addition, market researchers estimated that the drone market is estimated to increase by almost three times in 5 years from 2021-2026, which increases by USD 5.48 billion to 2026 to 15 billion USD and drones service markets is projected to grow from 13.9 to 40.7 billion USD by 2021-2026 with Compound Annual Growth rate of 23.8 percent from 2021-2030 as shown in Fig 1 (Anonymous.2022). The application of drones in agriculture provides great potential and hence it is important to study drones from an Indian agricultural point of view.

2. BENEFITS OF DRONE IN AGRICULTURAL SECTOR

Drone can be used for plural numbers in the agricultural sector such as Fixed wing, Single Rotor/Helicopter, Quadcopters, Hexacopter and Octacopter are tabulated in Table 1, includes pesticides, soil and crop nutrient spraying, soil samples, farm animal monitoring, real -time aerial imagination and sensor data collection, and yield estimates (Anonymous., 2023a). Here are some of the major benefits of using drones in agriculture shown in Fig. 2:

Table 1: Different type of drones in agriculture

Type of drone	Description	Figure	References
Fixed Wing	When a vehicle reaches a particular speed, the lift-producing effect of fixed wings, which have stationary wings in the shape of aero foils, is needed.		Li *et al*., 2022
Single rotor / Helicopter	It consists of a single set of blades that rotate horizontally and are fixed to a central mast to provide lift and thrust.		Wang *et al*., 2019
Quadcopter	The quadcopters model consists of a central mast supporting a set of four horizontally rotating blades.	**Developed by Division of Agricultural Engineering ICAR-IARI, New Delhi, India**	Wang *et al*., 2020 Anonymous., 2023c

Hexacopter	The hexacopter featuring a central body frame with six horizontally rotating blades.	**Developed by Division of Agricultural Engineering ICAR-IARI, New Delhi, India**	Yallappa *et al*., 2017 Anonymous., 2023c
Octacopter	The octacopter distinguished by a central body frame supplemented by eight horizontally rotating blades.		Gaba., 2022

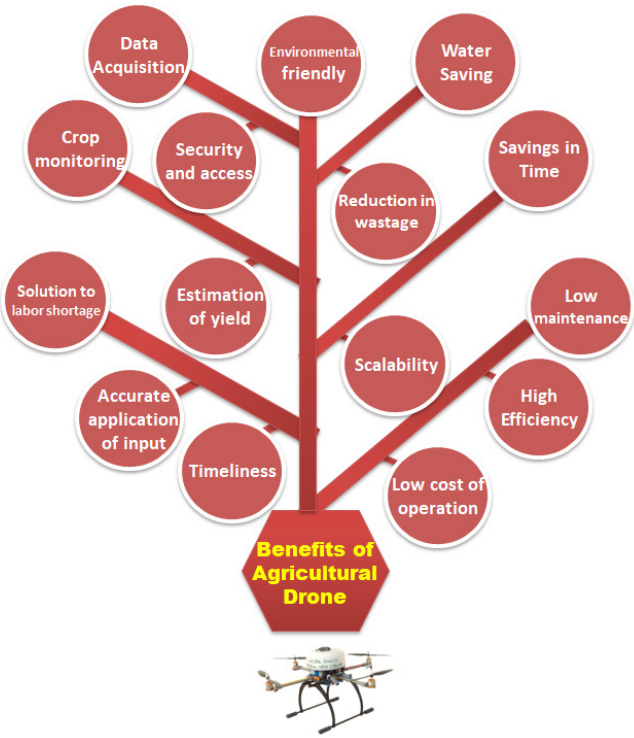

Fig. 2: Benefits of drone in agricultural sector

1. **Savings in time:** Drone can quickly and efficiently cover large areas, reduce the time required for traditional manual inspections.
2. **High efficiency:** Since the drone can be operated above the crop canopy, the drones can quickly and efficiently cover large areas. Increased crop yields, drones can be

used to collect data on crop health, allowing farmers to identify areas that require attention. The drone can spray 50–100 acres per day depending on the capacity of the drone, which is 30 times more than the traditional knapsack sprayer. By addressing these issues, farmers can improve their crop yields and increase their profits.

3. **Low cost of operation:** As drone crop can fly over the canopy, it can go to any point of the field without harassing the crop and fence. Drone can help reduce the cost by identifying the fields of the farm, which requires attention, requires to reduce the need for manual labor and reduce the use of pesticides and other chemicals.
4. **Accurate application of input:** Drone can apply input to a pinpoint place with the help of RTK GPS. This can help identify areas that require attention and help to ensure that intervention is targeted and effective.
5. **Estimation of yield:** With the use of imaging sensors, drones can capture data in the growing season, allowing farmers to analyze crop performance and predict potential yields.
6. **Security and access:** Monitoring crops and areas with drones eliminates the need to cross physically difficult or dangerous landscapes to farmers, such as steep areas or flood areas. This enhances safety and access by providing wide view of the entire region.
7. **Crop monitoring and management:** Drone equipped with high-resolution cameras and multisectoral sensors can capture wide images of crops from air. These images can be used to monitor crop health, detect diseases, pests, deficiencies of nutrients and identify areas with insufficient irrigation.
8. **Environmentally friendly:** As drones are capable of spraying chemicals at target locations, they can protect farmers and operators from dangerous chemicals. This helps to save the environment.
9. **Reduction in wastage:** Due to high level of nuclearization when using drones, about 30% of pesticides can be saved. Pesticides in the form of chemical fog can be sprayed at all levels of crop canopy.
10. **Saving water:** The capacity of a drone tank is usually limited to 10-15 liters. Therefore, high -centered chemicals are sprayed per unit area, resulting in 90% water saving compared to traditional spraying methods.
11. **Low maintenance:** Drones consist of less moving parts than traditional machines. They are rugged enough to use in the condition of agricultural rigid areas. The battery often requires charging and its life is limited.
12. **Sensor data acquisition and collection:** Many sensors can be applied on drones. Data can be easily collected from the sensor. Depending on the sensor data, the drone can decide in real time.

13. **Scalability:** Drone can be used with similar efficiency in small as well as large areas.
14. **Timeliness:** Agricultural drones are capable of performing spray operations without any time compared to traditional methods.
15. **Solve to the problem of labor shortage:** As a traditional method requires many laborers to spray pesticides, most of the time labor is not available on time. Drone provides a safe and fast service for spraying.

Challenges of adopting drone technology in the agriculture sector

The agricultural drone has immense potential and can offer many benefits to farmers, However, there are also some challenges and constraints that may be holding farmers back from adopting this technology. Here are some of the key challenges:

1. **High Initial Cost:** Cost: Most of the agricultural drones available for pesticide application with 10-15 kg payload capacity are costing around 6 to 10 lakhs depending upon their features. Additional sensors like terrain sensors, obstacle avoidance, etc.

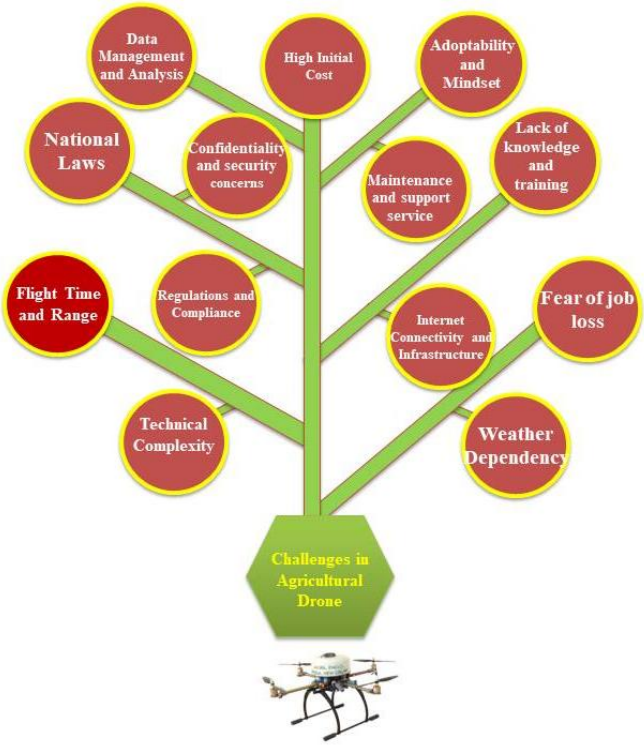

Fig. 3: Challenges of adopting drone technology in the agriculture sector

are costly and increase the overall cost of the drone. Small and marginal farmers may not prefer to buy the drone. However, they can get the service from CHC, FPO/ a company at an affordable rate.

2. **Lack of knowledge and training**: As the drone technology is new and is in the development phase, Farmers may not have the knowledge or training necessary to operate drones effectively. This can make it difficult for them to adopt this technology, as they may not be confident in their ability to use it.
3. **Fear of job loss:** Though the technology has the potential to attract the youth of rural areas, However, farmers are concerned that the adoption of drone technology will lead to job loss, as fewer workers will be needed to perform manual labour on the farm.
4. **Weather Dependency:** Drones are sensitive to adverse weather conditions, such as strong winds, rain, or extreme temperatures, which can impact their flight stability and safety. Unfavorable weather can disrupt planned drone flights, affecting data collection schedules.
5. **Internet Connectivity and Infrastructure:** Drone requires a reliable internet connection to broadcast data and get real-time updates. However, many rural areas, where agriculture is headed, lacks strong internet infrastructure, hinders the spontaneous communication between drones and farmers.
6. **Regulations and Compliance:** The use of drones in agriculture is subject to various rules imposed by civil aviation officers. Getting the necessary permits and following the rules of airspace can be a bureaucracy and time -taking process.
7. **Technical Complexity:** Running drones and processing the data collected by them requires technical skills and training. Many farmers, especially in rural areas, may not have the necessary expertise to operate and maintain the data collected or explain the data collected by them.
8. **Flight Time and Range:** With profit, there are some limitations of drones for agricultural use. Due to the relatively high payload, the flight duration of drones used in agriculture is less than 15–25 minutes. This provides limited coverage of land with every charge. The cost of drone increases significantly over a long flight time.
9. **National Laws:** Piloting the Drone requires obtaining a permit (UAOP) of unmanned aircraft operator, no permission for each flight through online digital sky platform for no permission (NPNT) compliance. The current digital sky platform only accepts registration and the API is not yet completely open
10. **Data Management and Analysis:** Drones produce large amounts of data, and it can be challenging to process and analyze this data effectively. Farmers require access to suitable software and equipment to make an understanding of data and achieve actionable insights from it.

11. **Confidentiality and security concerns:** Drone equipped with cameras and sensors increases privacy concerns for individuals living near the farm. There is a need to address the issues of privacy and establish guidelines to ensure that the data collected by the drone is safely handled to protect sensitive information.
12. **Adoptability and Mindset**: Some farmers may be resistant to changes and hesitation to adopt new techniques like drones. Explaining the benefits of drone technology and farmers of benefits may require educational and awareness efforts.
13. **Maintenance and support service**: Drones require regular maintenance and sometimes repair. It is important to ensure reliable technical assistance and access to spare parts to reduce downtime and disruption in drone operations.

To address these challenges requires collaborative efforts from various stakeholders, including government officials, technology providers, agricultural extension services and research institutes. By developing target aid programs, providing training and education and streamlining regulatory processes, the agricultural sector can overcome these challenges and is still in its early stages to fully redeem the ability of drone technology to change the practices of farming and increase productivity and adopt drone technology in agricultural sector in rural India. While there is interest in this technology, there are also concerns about job loss and lack of knowledge and training. However, efforts are underway to resolve these challenges and encourage drone technology.

One of the major initiatives is the Digital India campaign, which aims to provide digital infrastructure and connectivity in rural areas. This initiative focuses on training and education, which can help to overcome the lack of knowledge and training among farmers. In addition, there are many organizations and initiatives that focus on promoting the use of drone technology in agriculture. For example, the Indian Council of Agricultural Research (ICAR) has established a center for accurate and farming technologies, which focuses on promoting accurate agricultural technologies including drones.

Research issues in adopting agricultural drones in India

Adopting agricultural drones in India presents many research issues that require further investigation and investigation. Some of these researchers are involved in issues as shown in Fig. 4:

1. **Cost-Benefit Analysis:** It is necessary to conduct a comprehensive cost-profit analysis of adopting agricultural drones in India. Research should assess the overall economic viability of using drones in various crop types and agricultural sizes, considering initial investment, operational costs, labour savings and improving potential yield.
2. **Farmer Acceptance and Perception:** It is important to understand the attitude, perceptions and motivations of Indian farmers to adopt drone technology. Research

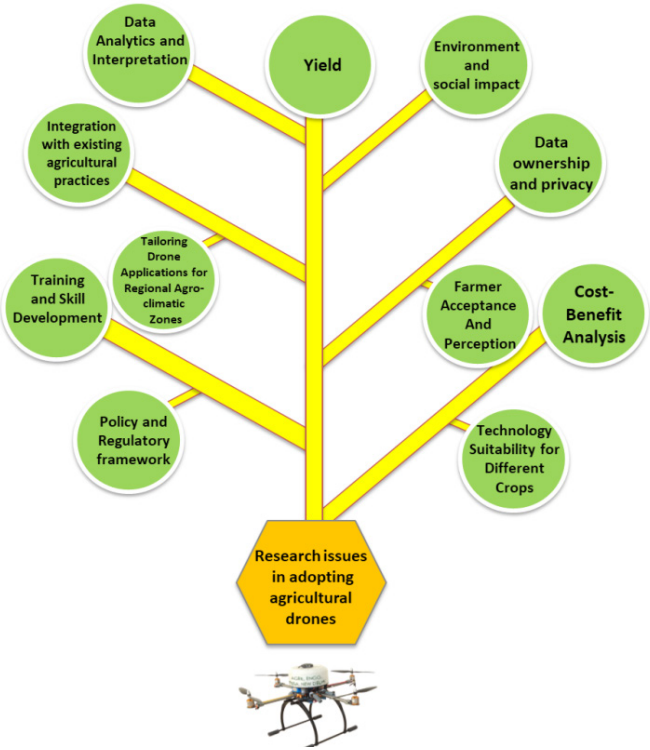

Fig. 4: Research issues in adopting agricultural drones in India

can detect factors affecting the desire to embrace this technique, alleged obstacles and to promote awareness in adopting education and awareness.

3. **Technology Suitability for Different Crops:** It is necessary to check the suitability and effectiveness of the drone for various crops grown in India. Different crops have separate growth patterns and requirements, and research can focus on identifying which crops benefit the most from drone applications and which cannot hold investment right.
4. **Tailoring Drone Applications for Regional Agro-climatic Zones:** There are various agricultural climatic areas in India, which have different farming practices. Research should check how drone techniques can be adapted and optimized to suit the specific requirements of different regions, crops and farming systems.
5. **Yield:** It is important to assess the actual effect of adopting drones on agricultural productivity and yield. Long -term studies can help determine whether the use of drones leads to frequent and permanent improvement in agricultural production.
6. **Data Analytics and Interpretation:** Research can focus on improving data analytics and interpretation methods for drone-ecastic data. Developing advanced algorithms

and machine learning models can help achieve actionable insights from data and make them more accessible to farmers.

7. **Integration with existing agricultural practices:** Understanding how drones can be basically integrated with traditional agricultural practices, it is necessary for successful adoption. Research can find out how drones can complement the existing equipment and techniques used by Indian farmers.
8. **Training and Skill Development:** Drone technology related training programs and effectiveness of the initiative of capacity building are important. Research can effectively assess the impact of training on farmers' capacity to operate and maintain drones.
9. **Policy and Regulatory framework:** Analysis of existing policies and rules related to drone use in agriculture is important to identify areas for potential obstacles and improvements. Research may recommend policy changes to promote adoption of agricultural drones, ensuring safety and compliance.
10. **Environment and social impact:** Research on the environment and social effects of adopting drones in agriculture is necessary to understand the widespread implications of this technique. Using pesticides, water consumption and its ability to reduce carbon footprint can be important in promoting permanent practices.
11. **Data ownership and privacy:** It is important to examine issues related to data ownership and privacy in agricultural drone use. Research can find out how farmers 'data is stored, stored and used by drone service providers and how to protect farmers' data rights.

Addressing the issues of these researchers to promote effective and responsible adoption of agricultural drones in India, eventually providing valuable insight to the farmers and agricultural sector overall and can help develop a strategy.

3. CONCLUSION

Drone provides several benefits to farmers in the agricultural sector, including increased efficiency, better yields and low costs. However, there are also concerns about job loss and lack of knowledge and training that can withdraw farmers by adopting this technique. Nevertheless, adopting drone technology in rural India is still in its early stage, efforts are underway to resolve these challenges and promote the use of drones in agriculture. It is important for farmers to understand the potential benefits of this technique and get the training and support necessary to use it effectively. At conclusion, adopting drone technology in agriculture has the ability to manage their crops and improve their yields. While there are challenges to overcome, such as job loss and concerns about lack of knowledge and training, initiatives are also underway to provide assistance to farmers who wish to promote the use of this technology and adopt it. It is necessary that farmers and policy makers work together to ensure that the

benefits of drone technology in agriculture are realized, while any concern or challenges are facing. By doing this, we can help create a more durable and productive agricultural sector that equally benefit both farmers and consumers.

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